

Vectors and Vector-Valued Functions



Vector operations in 3D

- 1 Let $\mathbf{a} = \langle 2, -1, 3 \rangle$ and $\mathbf{b} = \langle 1, 4, -2 \rangle$. Find:

a $\mathbf{a} + \mathbf{b}$

b $\mathbf{a} \cdot \mathbf{b}$

c $|\mathbf{a}|$

- 2 Find the cross product $\mathbf{a} \times \mathbf{b}$ for $\mathbf{a} = \langle 1, 0, 2 \rangle$ and $\mathbf{b} = \langle 3, 1, -1 \rangle$.
-

Vector-valued functions: velocity and acceleration

- 3 A particle's position is $\mathbf{r}(t) = \langle t^2, 3t, t^3 \rangle$.

a Find the velocity vector $\mathbf{v}(t)$.

b Find the acceleration vector $\mathbf{a}(t)$.

c Find the speed $|\mathbf{v}(1)|$.

Unit tangent vectors and arc length

- 4 For $\mathbf{r}(t) = \langle 3\cos t, 3\sin t, 4t \rangle$, find $\mathbf{r}'(t)$ and the speed $|\mathbf{r}'(t)|$.

- 5 Find the arc length of $\mathbf{r}(t) = \langle 3\cos t, 3\sin t, 4t \rangle$ for $0 \leq t \leq 2\pi$, using the constant speed found above.
-

Applications of the cross product

- 6 Find a vector orthogonal to both $\mathbf{u} = \langle 1, 2, 0 \rangle$ and $\mathbf{v} = \langle 0, 1, 3 \rangle$, using the cross product.

- 7 Find the area of the parallelogram spanned by $\mathbf{u} = \langle 2, 0, 0 \rangle$ and $\mathbf{v} = \langle 1, 3, 0 \rangle$, using $|\mathbf{u} \times \mathbf{v}|$.
-

Lines and planes in space

- 8 Write parametric equations for the line through $(1, 2, -1)$ in the direction of $\mathbf{v} = \langle 3, -1, 2 \rangle$.

- 9 Find the equation of the plane through $(1, 0, 2)$ with normal vector $\mathbf{n} = \langle 2, -3, 1 \rangle$.
-

Motion along a curve

10 A particle's position is $\mathbf{r}(t) = \langle e^t, e^{-t}, \sqrt{2}t \rangle$. Find $\mathbf{v}(t)$ and show that the speed $|\mathbf{v}(t)|$ is minimized at $t = 0$.